

Insulin Dose Calculator — Parameter Grid (Ranges)

ORDER PARAMETERS — ALLOWED RANGES

Parameter	U-100 Standard (Units)	U-10 Dilute (Syringe Units)
Insulin : carb ratio	1–100 g/Unit	1.5–40 g/Syr Unit
Correction factor (ISF)	4–300 mg/dL/Unit (verify > 150; high warn ≥ 200)	10–200 mg/dL/Syr Unit
Target blood glucose	120–300 mg/dL	120–300 mg/dL
Rounding increment	0.5 or 1 Unit (prescriber-set)	1 Syr Unit (fixed)

ORDER-SIGN SOFT ADVISORIES — BY AGE

Age band	U-100 Standard only		Both U-100 & U-10	
	I:C low (g/Unit)	I:C high (g/Unit)	CF low (mg/dL/Unit)	Target BG (mg/dL)
0–5 y	below 10	100 or more (all ages)	below 50	under 130
5–11 y	below 4		below 15	over 250
12 y and older	below 2		—	over 250

CALCULATOR PARAMETERS — ENTRY GUARDS

Parameter / input	U-100	U-10
Target BG · < 5 y	130–300 mg/dL	130–300 mg/dL
Target BG · 5–11 y	120–250 mg/dL	—
Target BG · 12 y and older	120–250 mg/dL	—
Carbohydrates eaten	0–300 g; Alerted Calculation cap of 250 g	0–99 g
Current blood glucose	1–2000 mg/dL; alerted cap at 600	1–2000 mg/dL; alerted cap at 600

BLOOD-GLUCOSE DECISION THRESHOLDS (both concentrations)

Condition	Behavior
BG = 0	Critical hypoglycemia — functionally never triggered (entry minimum is 1)
BG < 70	Hypoglycemia — all dosing suppressed
BG > target	Correction applies (otherwise prandial only)
BG > 600	Correction calculated at 600 cap; prandial unaffected

- Order details are hard limited at “add to scratch pad” and further refined via soft advisory warnings at “sign order” when detail values for ICR, CF, or target BG are allowed but outside common values for age groups.
- Target BG ranges are age tiered and concentration-independent: the same soft order-warn (on sign) and calculator limits apply to U-100 and U-10 at each tier.
- The I:C high advisory fires at 100 (the maximum), flagging attempts to use the calculator for sliding-scale-style orders where a very high ratio makes prandial coverage negligible.
- CF carries two graduated high cues at sign: verify above 150 and a dedicated high warn at 200 mg/dL/Unit (all ages).
- Ordered values that are outside soft warning limits are re-checked at the calculator.
- BG entry is 1–2000 mg/dL; correction uses up to 600 (alerted). BG = 0 cannot be entered, so critical hypoglycemia is functionally never triggered at the calculator.
- Out-of-range parameters are corrected up or down to the nearest violated boundary.
- Reference only — not for clinical use without pharmacy & informatics committee review.